

Ultra-low Noise Microwave Generation Using Femtosecond Lasers and Applications

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Abstract: We report on our fiber-based mode-locked laser approach to the generation of ultra-low noise microwave signals, and its use for frequency metrology. We make use of two erbium fiber mode-locked lasers phase locked to a common ultra stable laser source to generate an 11.55 GHz signal with a relative phase noise of -111 dBc/Hz at 1 Hz from the carrier. The corresponding residual frequency instability is below 2.3×10^{-16} at 1 s and about 4×10^{-19} at 6.5×10^4 s [1]. For a single fiber system, we demonstrate absolute stability about 3×10^{-15} between 1 s and 10 s when locked to an ultra-stable optical cavity and after dissemination over amplitude-modulation fiber link for a few hundred meters. The microwave signal from the fiber system is subsequently used to perform Ramsey spectroscopy in a state-of-the-art Cesium fountain [2]. The resulting clock system exhibits a stability of $3.5 \times 10^{-14} \tau^{-1/2}$ limited by projection noise. These experiments demonstrate the potentiality of ultra-high-performance, robust and long term operations fiber-based frequency combs for operational-level tasks in time and frequency metrology using microwave signals.

1. Microwave extraction from an ultra-stable optical reference with a fiber-based optical frequency comb

In the line of previous work at NIST [3,4] and PTB [5], we present first our latest results on microwave extraction with fiber-based optical frequency combs. Two separate femto-second fiber lasers are optically-mixed with the same ultra-stable optical reference cw laser at $1.55 \mu\text{m}$ to provide beatnotes (f_b). The carrier envelop offsets (f_0) of the two lasers are independently measured using a standard f-2f technique. By mixing f_0 and f_b and mixing with a RF reference, we derive error signals for each laser which are retro-acted on the pump powers. This effectively locks the femtosecond lasers on the cw reference. The two pulsed lasers therefore exhibit ultra-stable repetition rates (down-converted versions of the cw-reference's optical frequency) which can be chosen to be identical and therefore easily compared. The repetition rates harmonics near 11.55 GHz are isolated, amplified and phase compared using a standard mixer-based demodulation technique. The resulting phase noise, long-term timing drift and corresponding Allan variance are presented on figure 1.

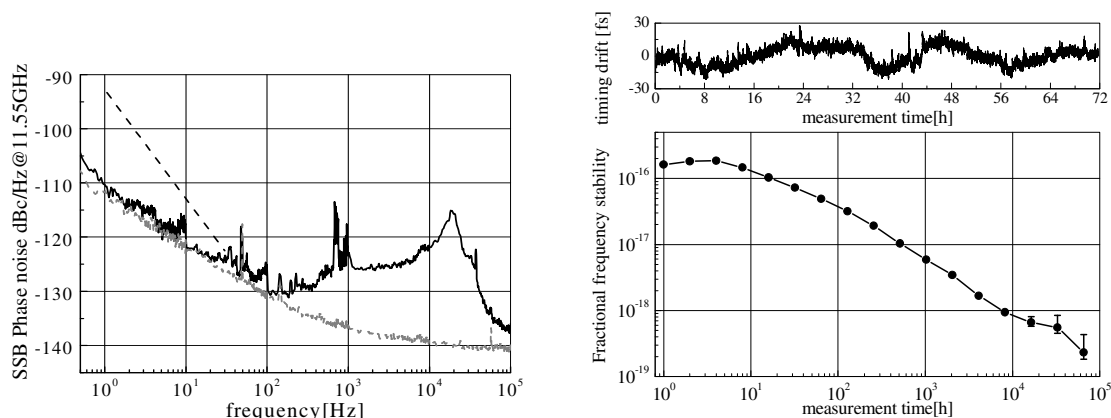


Figure 1 : Left : Single sideband (SSB) phase noise at 11.55GHz carrier of two microwave signals extracted from the same optical reference by two quasi identical fiber-based frequency combs. In gray : the measurement noise floor of the readout system. Right : corresponding long-term timing drift and Allan variance for a single microwave extraction system.

The measured residual phase noise at 1Hz from the 11.55 GHz carrier is -111 dBc/Hz (limited by the readout system). The corresponding frequency stability for a single system is about 1.6×10^{-16} at 1s. It averages down to about 3×10^{-19} at 65536 s, with a frequency error compatible with zero within the error bar. This is, to our

knowledge, the best results reported, so far, for microwave extraction using optical frequency combs, for time-scale slower than 100 Hz. We also emphasize the reliability and robustness of our fiber-based system, which allows continuous operation over several days without interruption.

2. Use as local oscillator for an atomic fountain clock

The previous section's results were characterizing the microwave extraction process and were obtained with a single (common-mode) ultra-stable cw optical reference. To characterize the *absolute* microwave generation, we did two separate experiments. In a first experiment, we generated independently a microwave signal from a Titanium Sapphire (Ti:S) based femtosecond laser locked (using the same technique) on a separate ultra-stable optical reference [6] at 1062.5 nm. As the Ti:S apparatus lies in a separate room about 30 m away, in order to realize a direct comparison, the microwave signal generated by the fiber system was transmitted to this room by an amplitude modulation fiber link implementing a round-trip compensation scheme similar to [7]. In a second experiment, we transmitted the fiber-system microwave signal to a third room about 300 m away, where we compared it to the signal from a cryogenic Sapphire Oscillator (CSO) at liquid helium temperature [8].

Both comparisons exhibit a phase noise about $10^{-9}/f^2$ [rad²/Hz] (at 9.2GHz carrier) for Fourier frequencies f in the 0.1Hz-10Hz range and an Allan deviation about 3×10^{-15} at 1 s integration time. These performances, among the very best for microwave sources, along with the reliability and robustness of the fiber-based system qualifies it as an excellent microwave source for long-term operation of atomic fountain clocks.

In a last experiment, we used our fiber-based system to generate a microwave signal at 11.932 GHz, directly suitable to drive the frequency chain of the LNE-SYRTE's FO2 atomic fountain clock (which lies in the same room as the CSO). The 9.2 GHz signal produced by the frequency chain was used to perform Ramsey spectroscopy on Cesium atoms in the FO2 fountain and locked to the atomic signal. The resulting primary standard was characterized by comparison with the CSO and exhibits a stability of $3.5 \times 10^{-14} \tau^{-1/2}$, limited by the quantum projection noise limit. These performances are identical to the usual operation, which uses the CSO as the local oscillator. Comparable results were obtained simultaneously in the group at PTB [9].

This demonstrates the usability of our fiber-based microwave generation system for operational-level tasks in time and frequency metrology, and in particular, as a future replacement for the CSO in the LNE-SYRTE's primary standards clock systems.

3. References

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